# Phase 1 report – Spectral gap as behavioral proxy

Verdict: **PASS**

## Setup

Contralateral inhibition archetype with drive $I = [1.5, 1.5]$, time constant $\tau = 1$, and sigmoid $f(x) = \frac{1}{1+\exp(-4(x-1))}$. The weight grid is $w_{12}, w_{21} \in [0, 5]$ with a 50 times 50 mesh ($\Delta w = 0.1$).

For each cell, three quantities are computed:

- the middle-branch fixed point of the scalar reduction $\rho_1 = f(I - w_{21} f(I - w_{12}\rho_1))$, i.e. the analytical analogue of the symmetric saddle in the asymmetric 2D system,
- the dominant real part $\mathrm{Re}(\lambda_1)$ of the Jacobian at that fixed point and the spectral gap $\Delta = \mathrm{Re}(\lambda_1) - \mathrm{Re}(\lambda_2)$,
- a simulation-based WTA verdict over $t \in [0, 50]$, starting from the mirror-image perturbations $\rho^* \pm (0.05, -0.05)$, with WTA declared when both runs end with $|\rho_1 - \rho_2| > 0.3$ **and** with opposite signs of $\rho_1 - \rho_2$. The sign-opposition clause tightens the plan's stated criterion (plan §3.3), which by its bare letter would also flag strongly skew single-FP regimes as WTA; the parenthetical in the same section ('the system commits to one population dominating [which population depends on the initial condition]') makes clear that bistable choice is the intended meaning, and the tightening is the faithful reading.

## Closed-form diagonal result

On the diagonal $w_{12} = w_{21} = w$, the linearization of the WC system at the symmetric fixed point reduces to a matrix with off-diagonal entries $-wg$, where $g = f'(I - w\rho^*)$ is the sigmoid slope at the fixed-point input. Eigenvalues are $(-1 \pm wg)/\tau$ and the pitchfork condition $wg = 1$ yields

$$w^* = 1.000008, \quad \rho^*(w^*) = 0.499998, \quad g(w^*) = 1.000000.$$

## Classification accuracy

Binarising the spectral predictor at the sign of $\mathrm{Re}(\lambda_1)$ (WTA when the dominant real part is positive) and comparing cell-by-cell against the simulation-based ground truth yields an accuracy of **99.96 %** (acceptance threshold: $\geq$ 95 %).

The analytical bifurcation curve was traced by bisection along radial slices in the positive quadrant (181 angles); its distance from the empirical boundary cells is summarized by

- median: 0.0527 weight units ($= 0.52$ grid cells),
- maximum: 0.1016 weight units ($= 1.00$ grid cells).

The plan's acceptance threshold is within one grid cell ($\Delta w = 0.10204081632653061$) at every boundary point.

The spectral gap changes sign (i.e. $\mathrm{Re}(\lambda_1)$ crosses zero) precisely at the empirical WTA boundary: **yes**.

## Figures

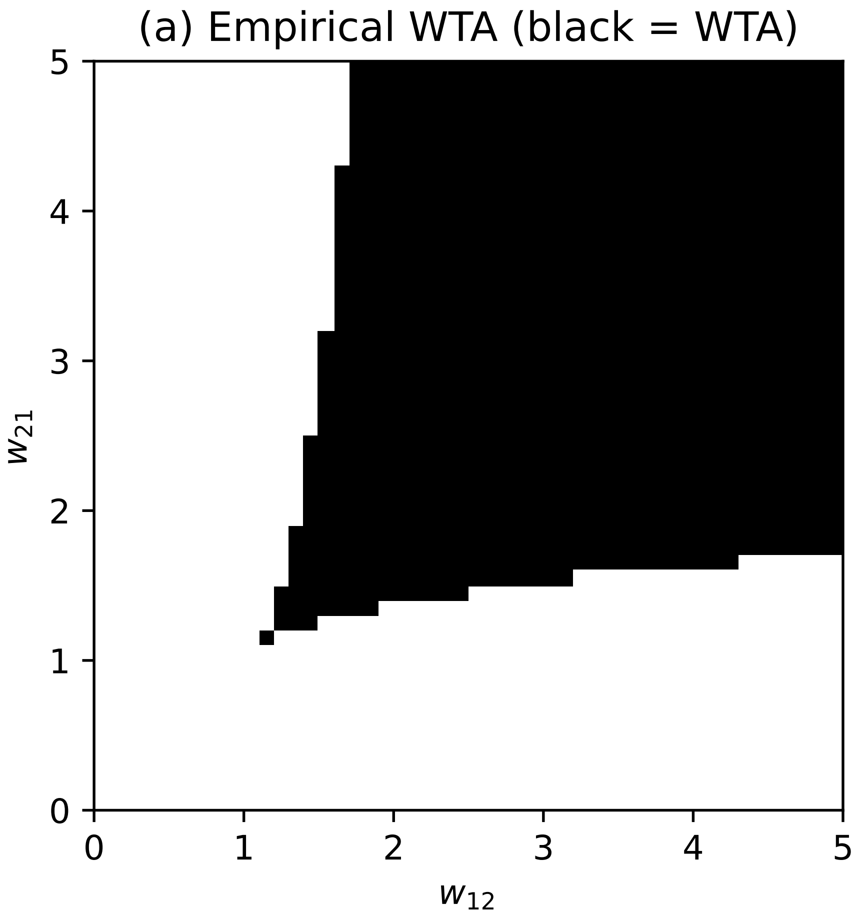


Figure 1: Panel (a). Empirical WTA map on the 50 times 50 weight grid. Black cells commit to winner-take-all from both symmetry-breaking initial conditions within $t = 50$; white cells remain symmetric.

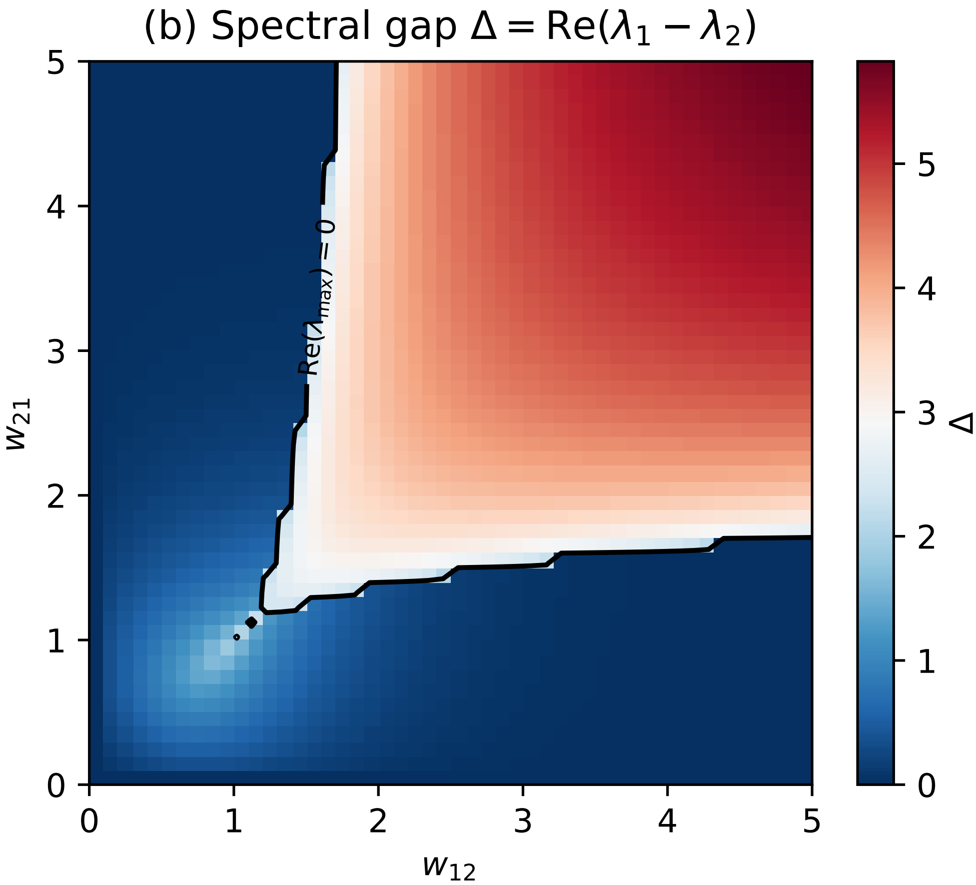


Figure 2: Panel (b). Spectral gap $\Delta = \mathrm{Re}(\lambda_1) - \mathrm{Re}(\lambda_2)$ at the symmetric fixed point. The black contour marks the zero-level of the dominant real part, i.e. the linear-stability boundary.

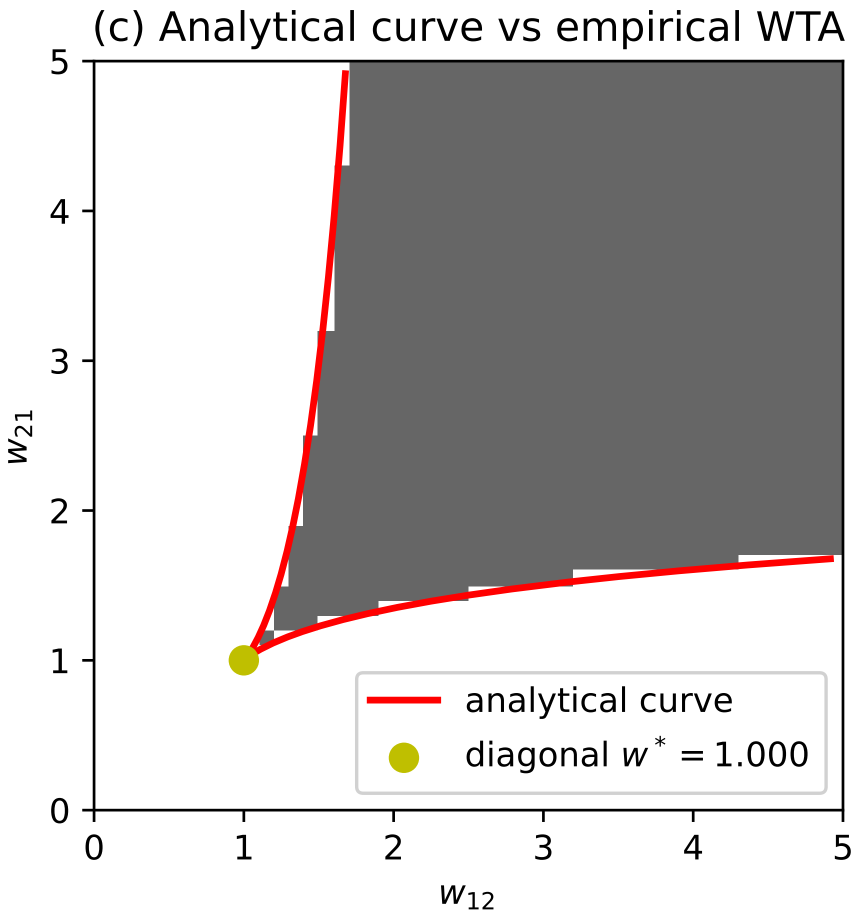


Figure 3: Panel (c). Fine-grained analytical bifurcation curve (red) obtained by bisection on the dominant eigenvalue, overlaid on the empirical WTA map (grey). The yellow marker locates the closed-form diagonal pitchfork point $w^* = w_{12} = w_{21}$.

## Verdict

Accuracy $\geq 95$ %: yes. Analytical curve within 1 grid cell at every boundary point: yes (max = 1.00 cells). Dominant real part crosses zero at the boundary: yes.

**Overall verdict: PASS.**